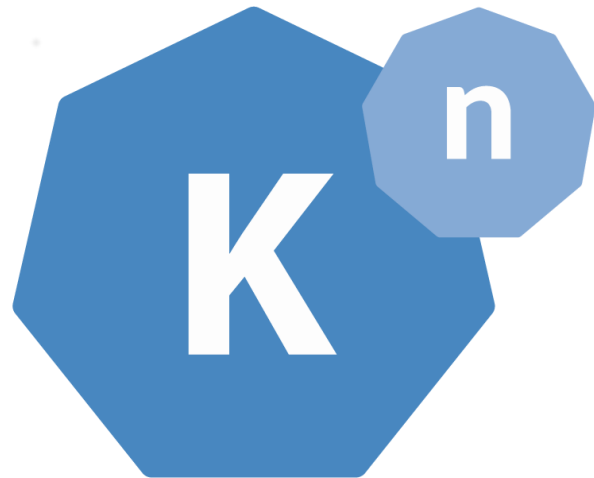




Knative Eventing

Bryan Zimmerman | Google

Nicolas Lopez | Google



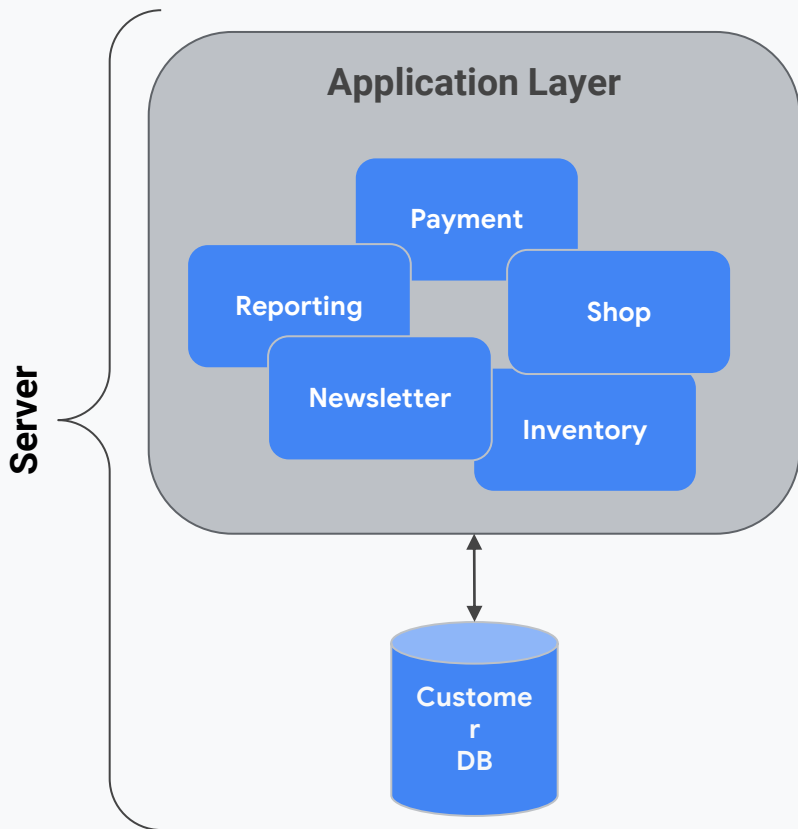
Agenda

- The Rise of MicroServices
- Event-driven Architecture
- Knative Eventing
- Demo: Image Processing
- Knative Community



The Rise of **Microservices**

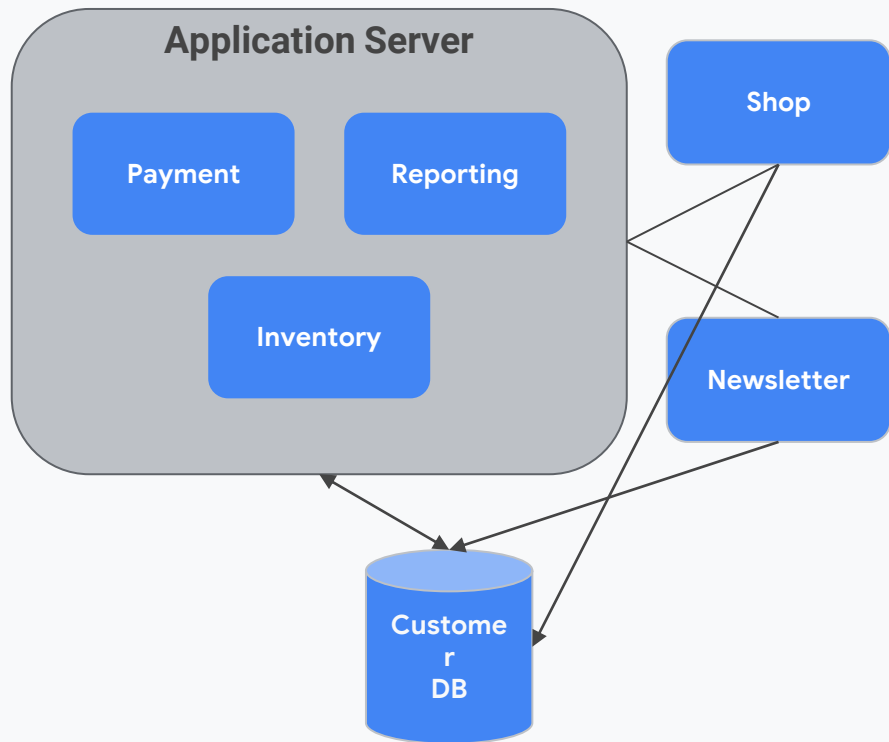
E-Commerce Store



● Monolith

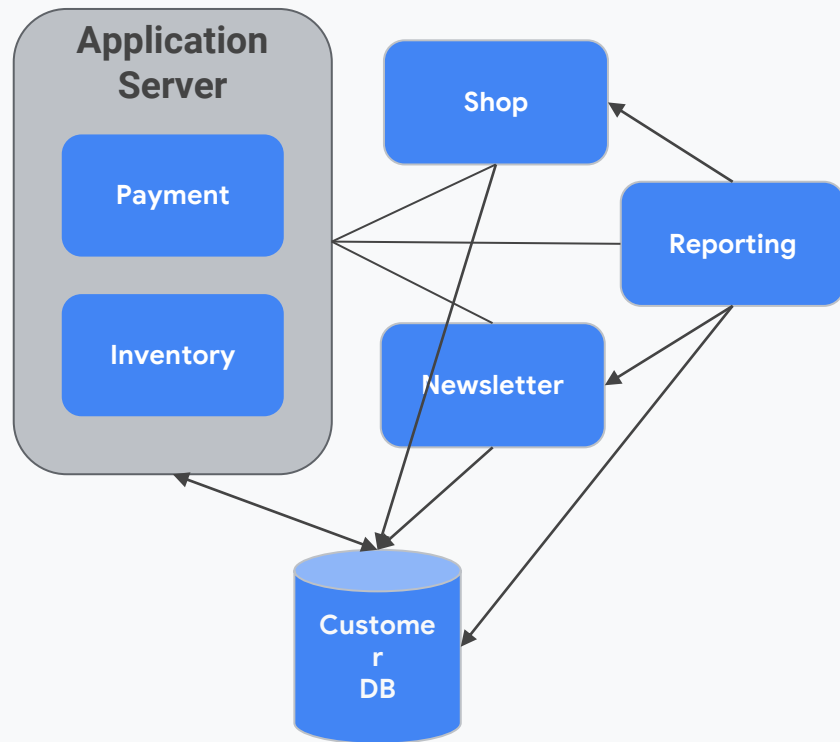
- Scales **Vertically** not Horizontally.
- **Overwhelming** to build, deploy and maintain as teams and application grow.
- **Nightmare** to change something.

E-Commerce Store



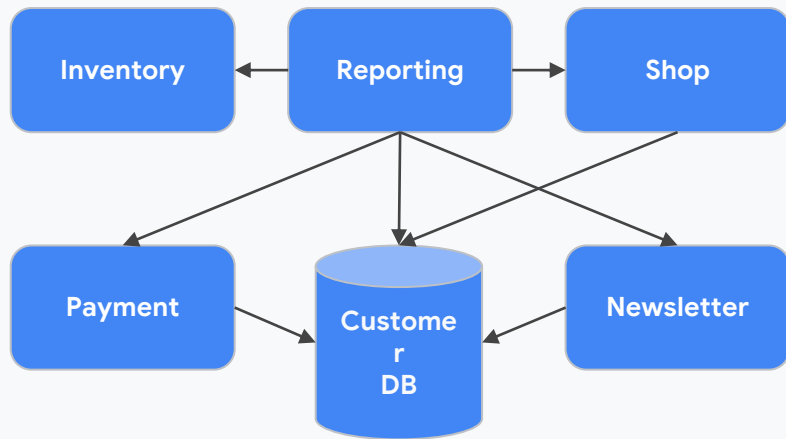
- Monolith
- Microservices

E-Commerce Store



- Monolith
- Microservices

E-Commerce Store

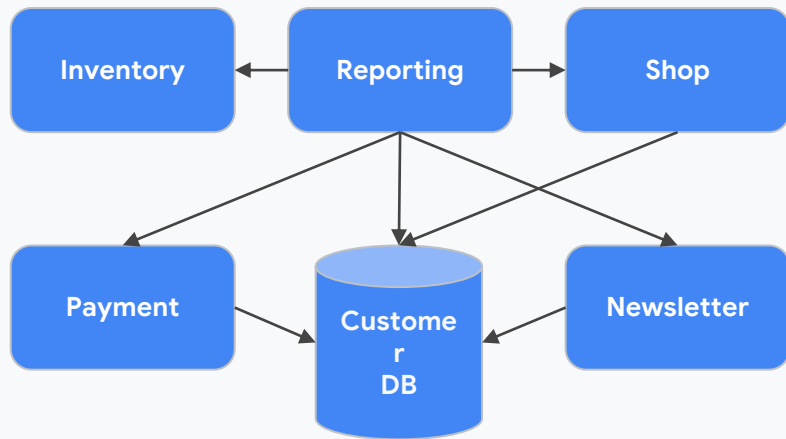


Monolith

Microservices

- "Separation of concerns"
 - Promotes Agile development.
 - Easy to build, maintain, and deploy.
 - Flexibility to build in the language of your choice.
- Services can be scaled independently to meet demand.

E-Commerce Store



● Monolith

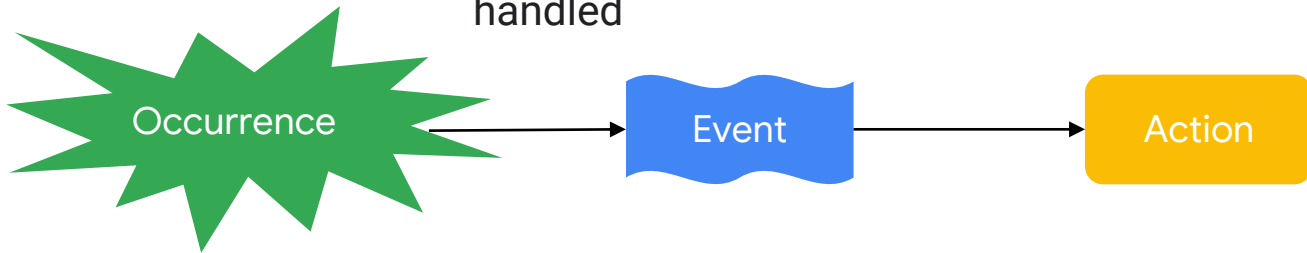
● Microservices

- Services can be **scaled independently** to meet demand.
- **"Separation of concerns"**
 - Promotes Agile development.
 - Easy to build, maintain, and deploy.
 - Flexibility to build in the language of your choice.
- **Spider web** of point-to-point integrations.
- **Coupling** with p2p inter-service communication.
- Adding/Removing services **requires changing** other services.

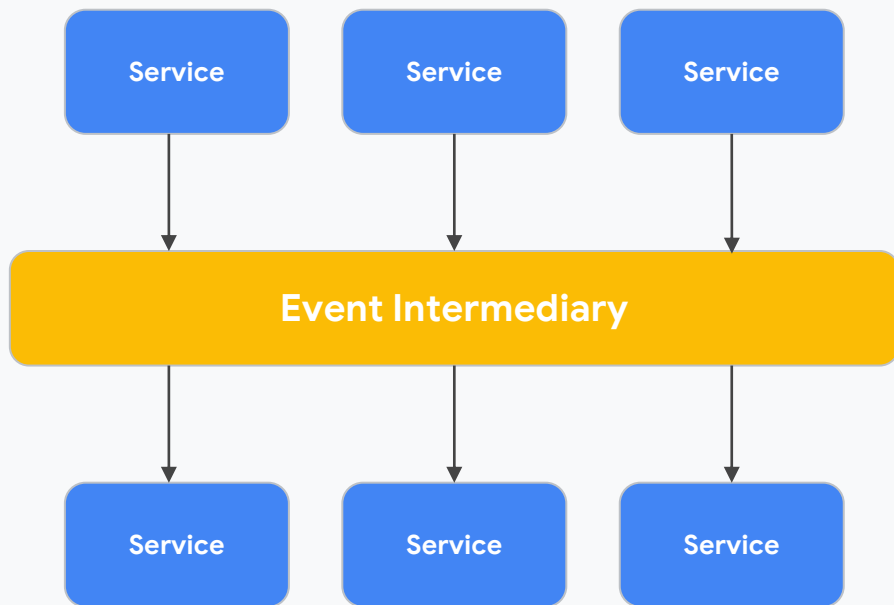
Event-Driven Microservices

What is an Event?

- An **event** is a record expressing an occurrence and its context
- Events represent facts and therefore do not include a destination
- The producer has no expectation of how the event is handled



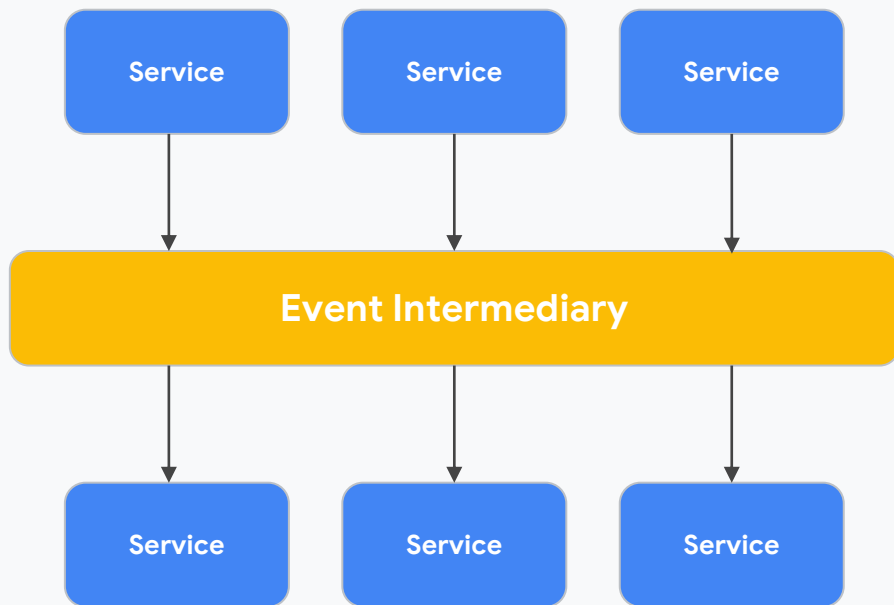
Event Producers



Event Consumers

Event-driven Microservices

Event Producers



Event Consumers

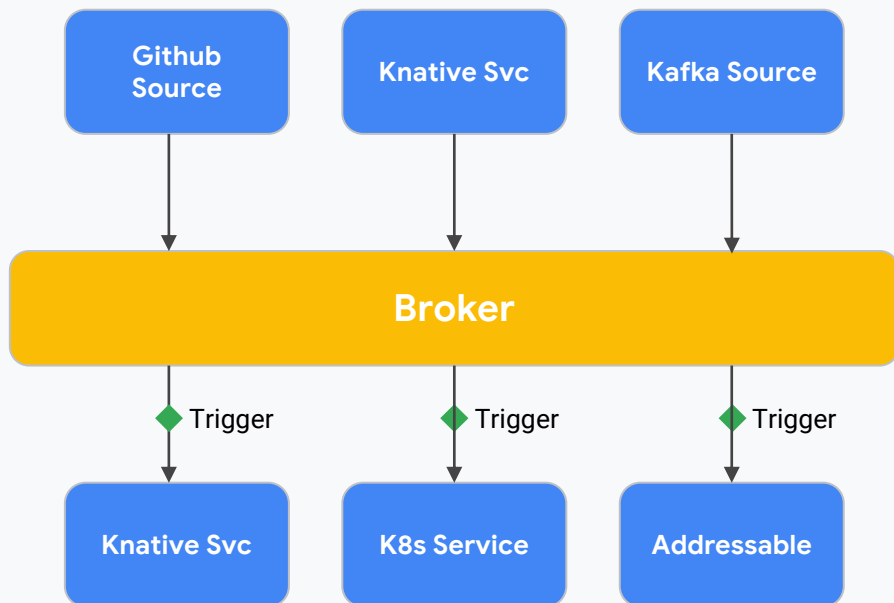
Event-driven Microservices

- Fully decoupled with no point-to-point communication.
- Highly Scalable.
- Extend organically - Add/Remove services without modifying existing application.



Knative Eventing

Event Producers



Event Consumers

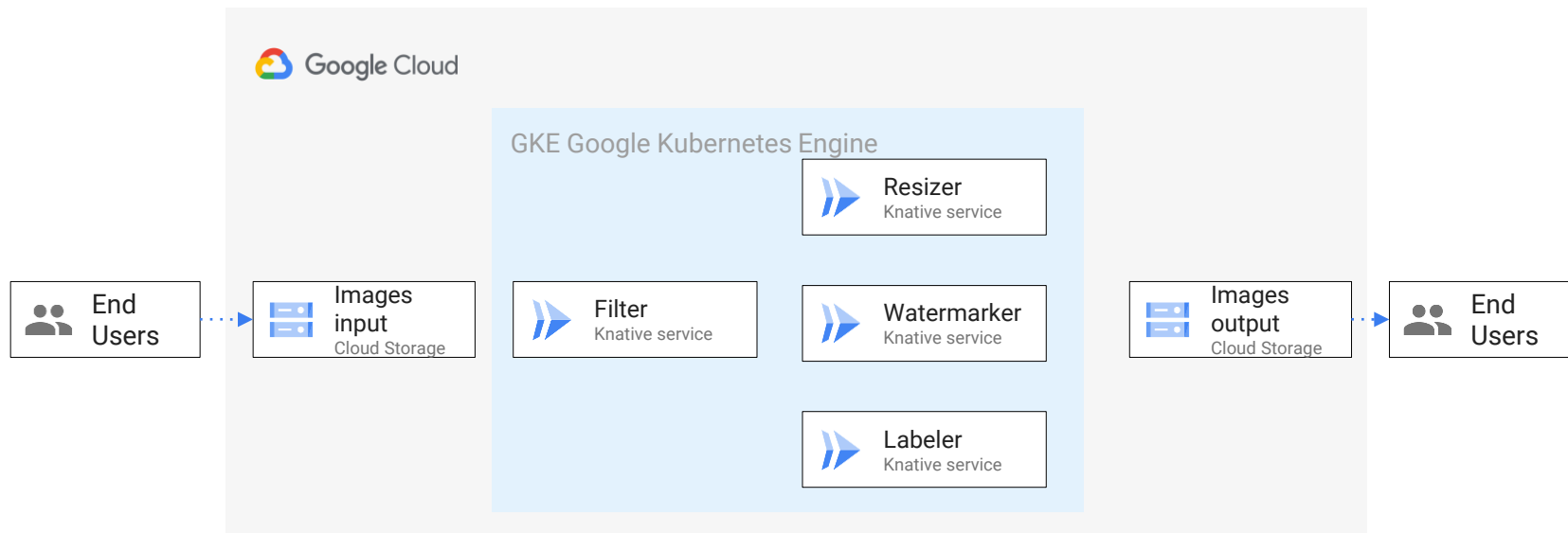
Knative Eventing

- Composable primitives to enable late-binding event producers and event consumers.
- **Cloud Events** to standardize event data
- **Event Producers (Sources)** produce Cloud Events
- **Event Consumer** is any Addressable such as Knative Service, K8s Service
- **Broker** and **Trigger** as intermediary
- Events and Source **Registry** for discovery.
- More primitives - Channels, Sequence, Parallel etc

Knative Eventing in Action

Image Processing Pipeline application

Image Processing Pipeline

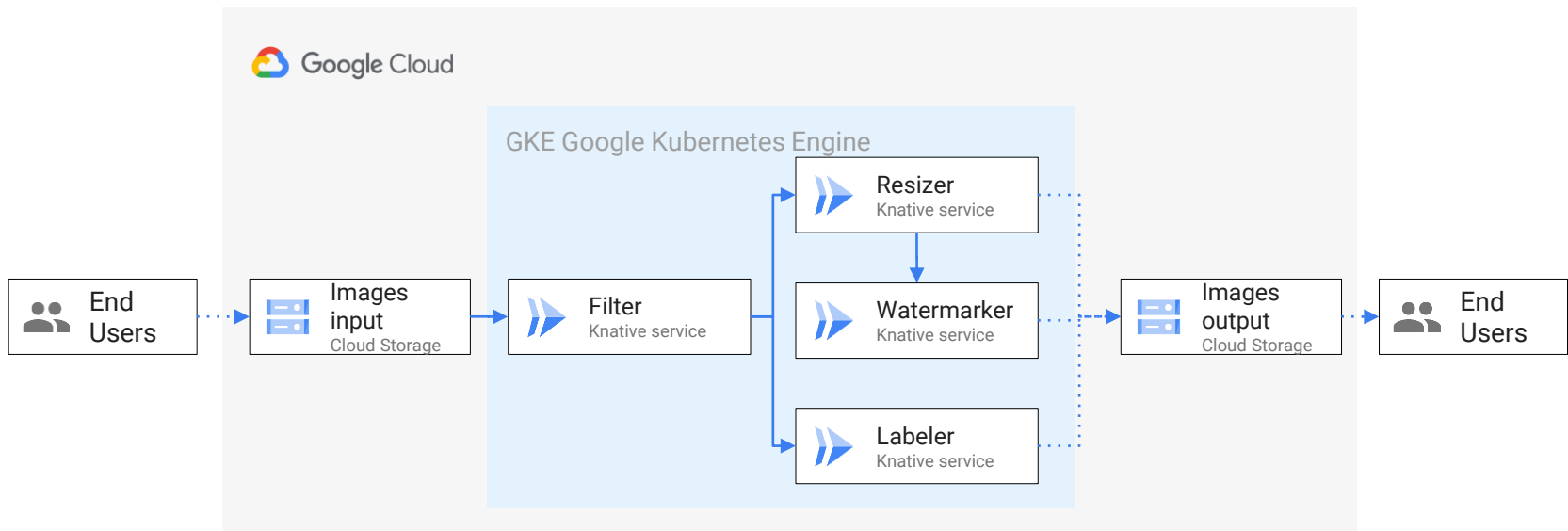


<https://github.com/meteatamel/cloudrun-tutorial/blob/master/docs/image-processing-pipeline-gke.md>

Image Processing Pipeline application

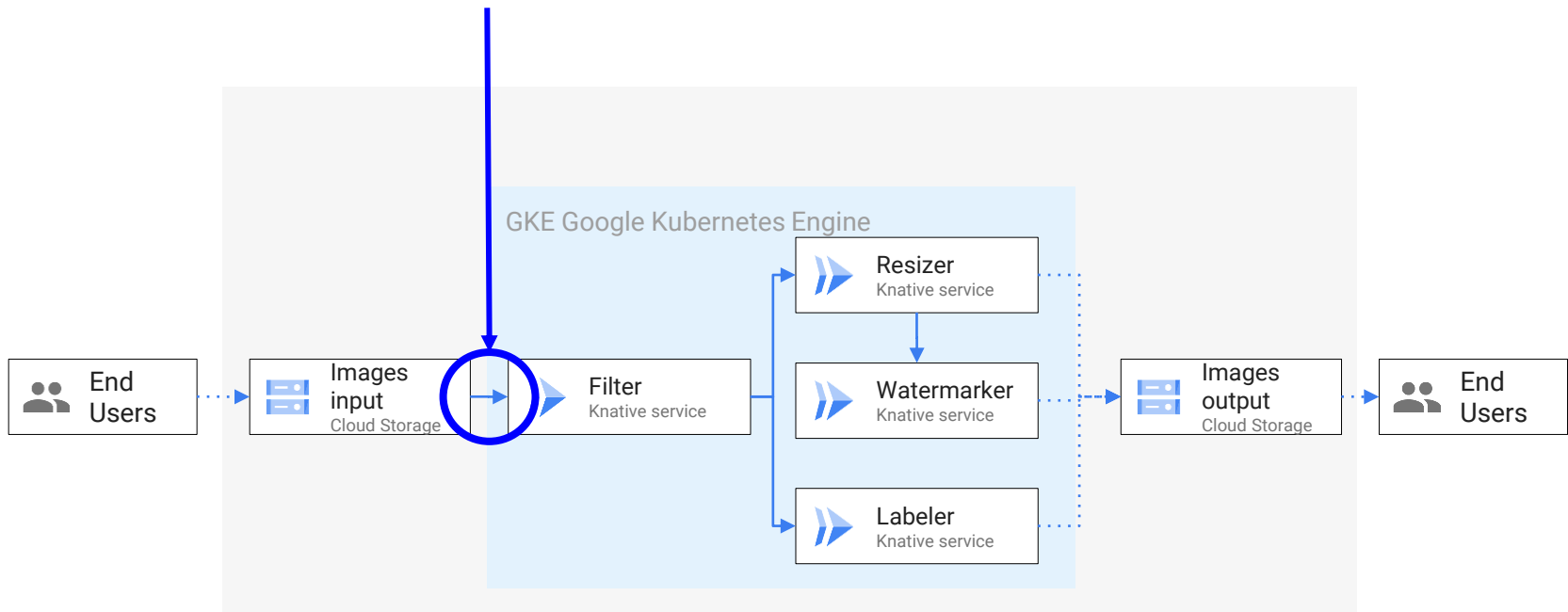
- We will setup **1 source + 4 triggers** to connect the services via events

Image Processing Pipeline



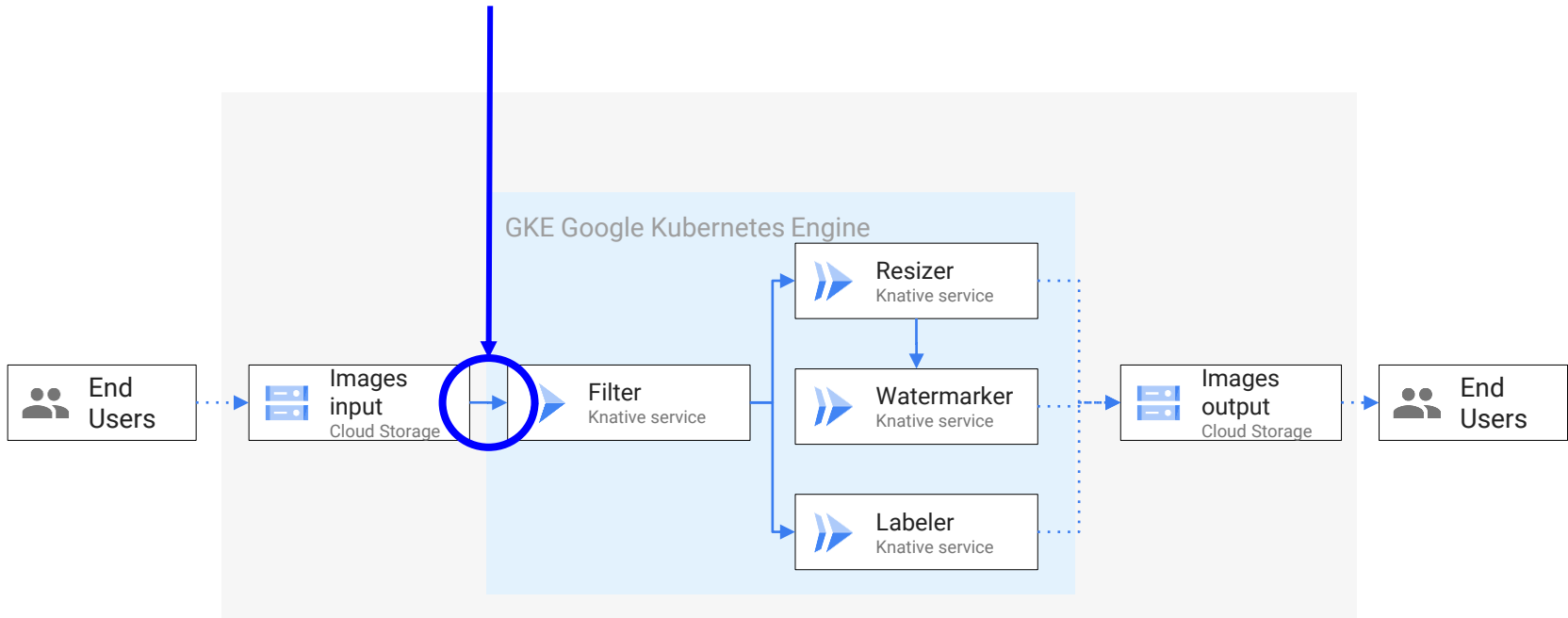
CloudStorageSource + trigger

- When an object gets stored in GCS, an event will be sent to the cluster



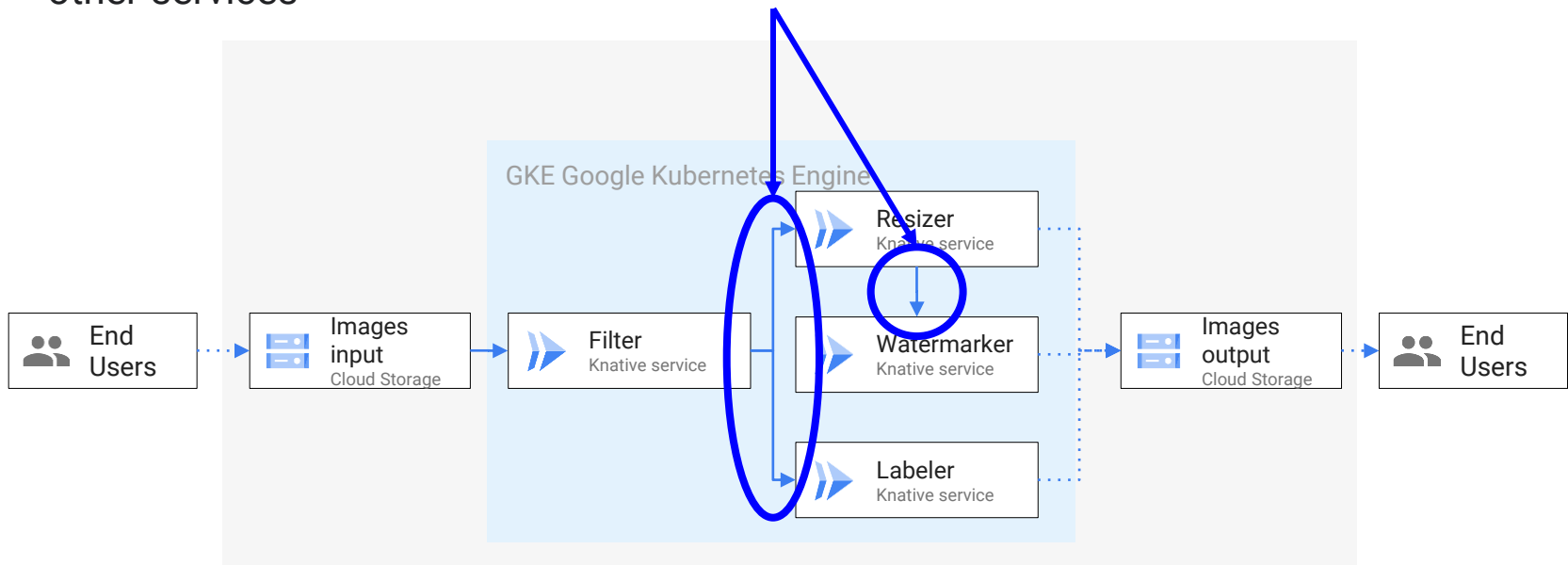
CloudStorageSource + trigger

- **Trigger** allows Storage Source events to be routed to the **Filter** service



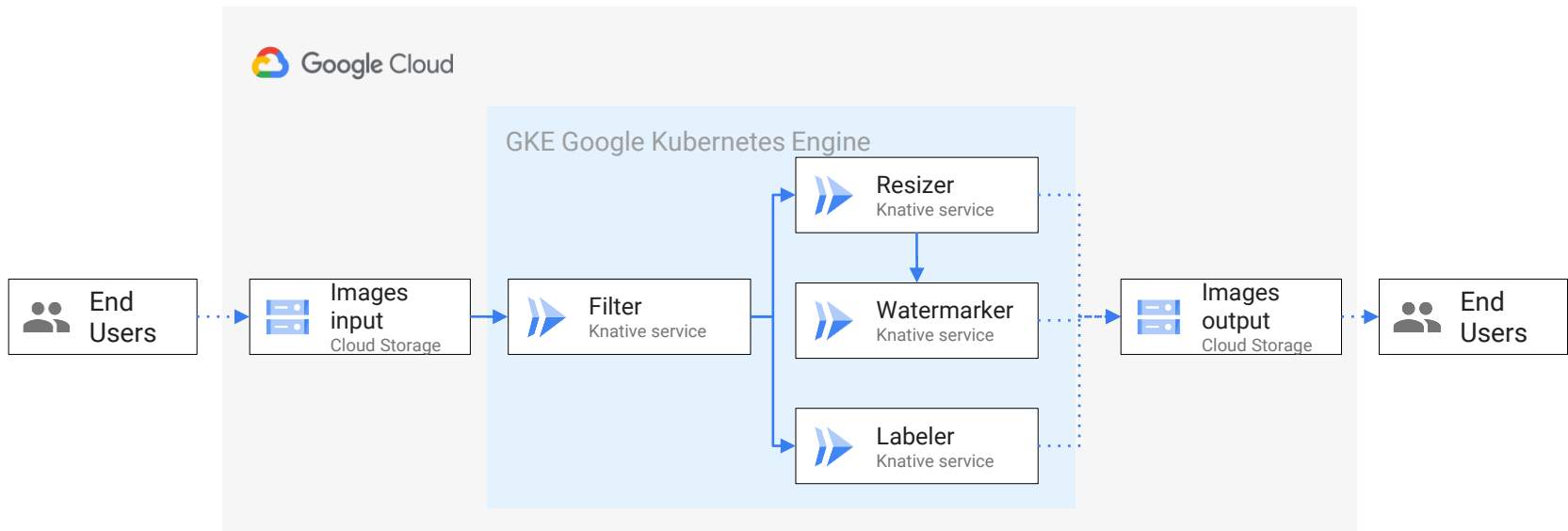
Custom Triggers

- Custom events are sent by the services with a **trigger** configured to route the events to other services



Ready!

Image Processing Pipeline



Knative Eventing Core Concepts

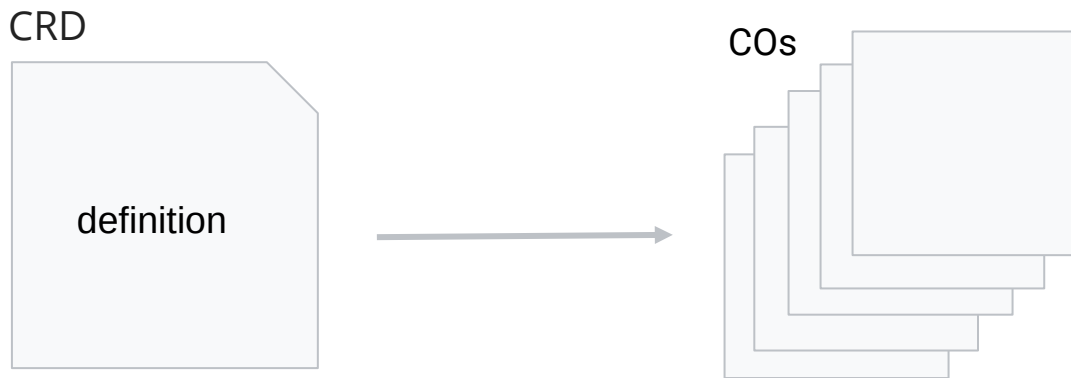
Knative Eventing Source

- An event Source
(<https://knative.dev/docs/eventing/sources/>) generates events or imports events from external producers
- Existing Sources
 - Community owned - Kafka, Github, etc.
 - Vendor owned
 - Google: CloudStorage, CloudAuditLog, etc.
 - TriggerMesh: AWS SQS, AWS Kinesis, etc.
 - Custom sources



How are Knative Sources implemented?

- Use standard way of extending Kubernetes: Custom Resource Definitions (CRDs)
- CRDs define new resource types → Class in a programming language
- Instances of CRDs are called Custom Objects (COs) → Object in a programming language



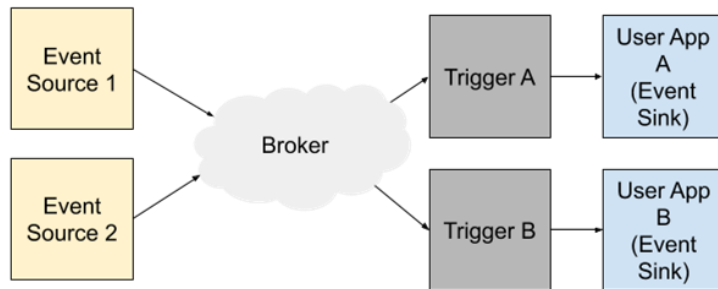
<https://kubernetes.io/docs/tasks/extend-kubernetes/custom-resources/custom-resource-definitions/>

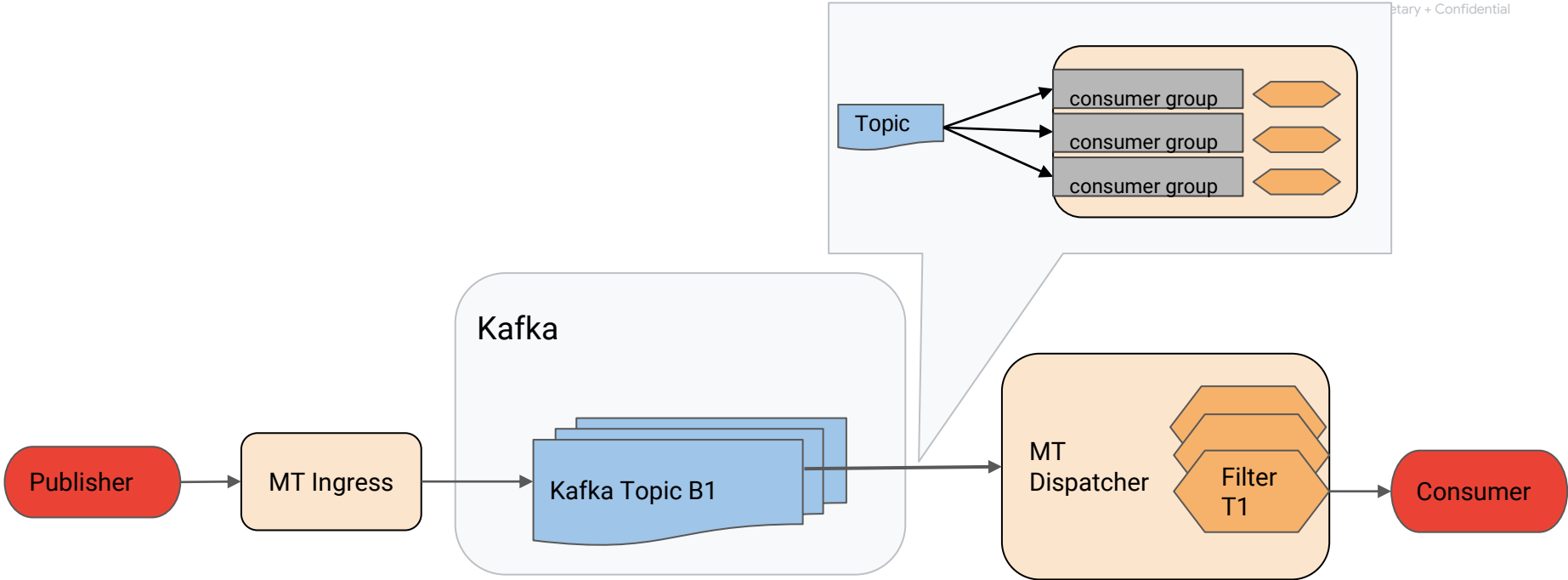
Types of Knative Sources

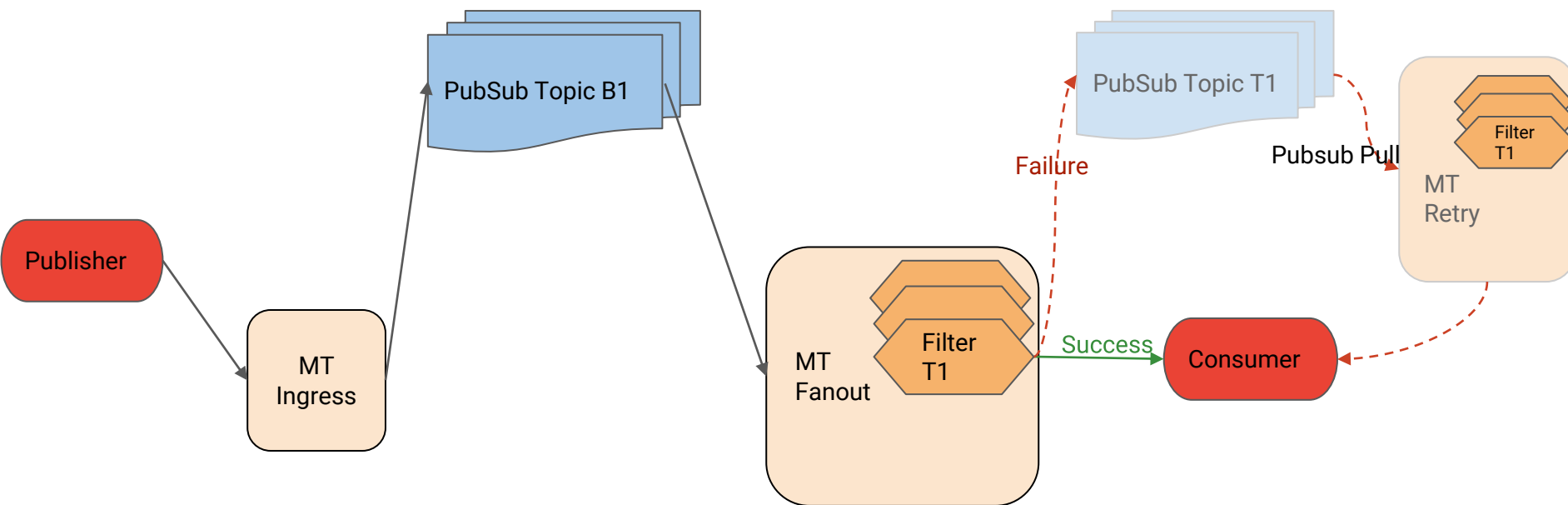
- Push
 - Must be exposed to the sender
 - E.g. a publicly exposed URL for GitHub to make requests against
 - Easier to scale
 - Can use Knative Services, scaling down to zero
- Pull
 - Something must always be running
 - Needs network access to the producer, but does not need to expose an endpoint
 - Harder to scale

Broker & Trigger

- Broker (<https://knative.dev/docs/eventing/broker/>)
- Triggers (<https://knative.dev/docs/eventing/triggers/>) are the way to subscribe events from a specific Broker
 - Filtering based on CloudEvents attributes
 - Event sink could be any “[addressable](#)”
- Broker implementations
 - Default
 - Alternative Broker implementations, e.g. GCP, Kafka

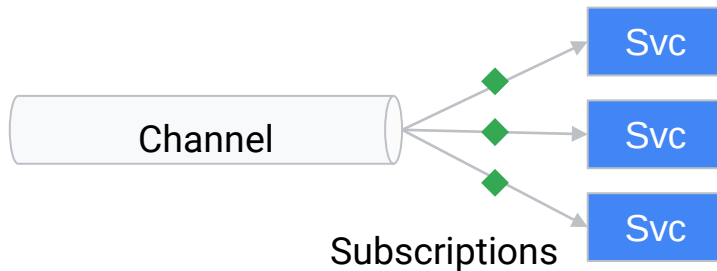






Primitives - Messaging

- Channel (<https://knative.dev/docs/eventing/channels/>)
- Subscription



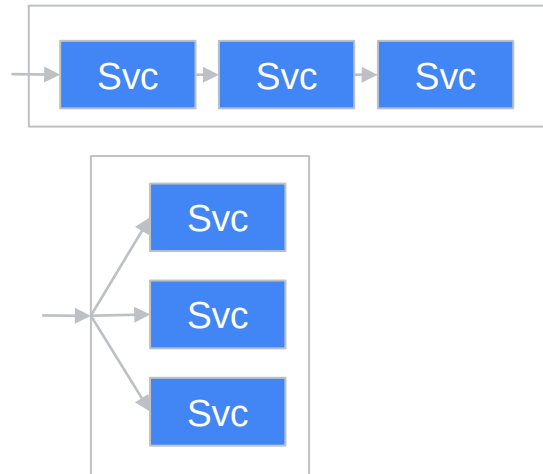
Primitives - Flows

- Sequence

(<https://knative.dev/docs/eventing/flows/sequence/>)

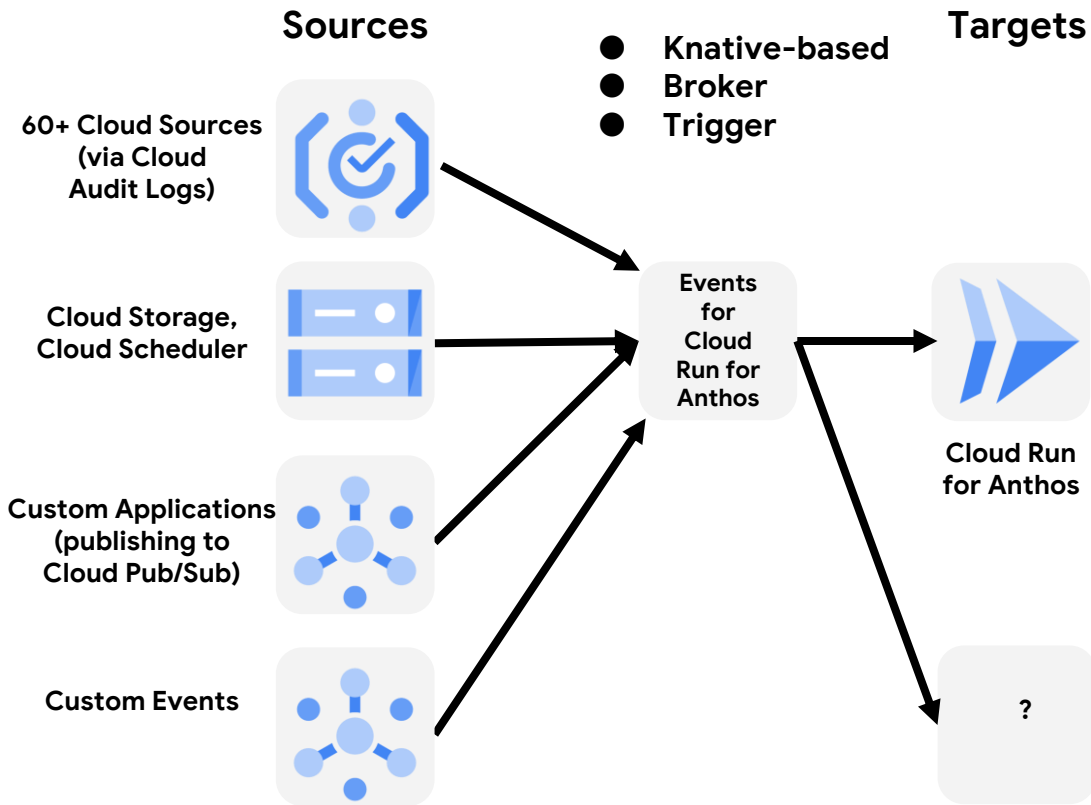
- Parallel

(<https://knative.dev/docs/eventing/flows/parallel/>)





Events for Cloud Run for Anthos



Knative Community

Vibrant community

Two core
components -
Serving and
Eventing

**10 active Working
Groups** with 450+
contributors

15+ active
repositories

**7 knative-based
offerings** from
vendors such as
Google, IBM, Red
Hat, TriggerMesh,
and VMWare.

Knative Partners



vmware®





What next?

Learn About Knative

<https://knative.dev/docs/>

Try our GCP implementation

<https://cloud.google.com/run/docs/events/anthos/quickstart>

View us on Github

<https://github.com/knative>

google_logo Open
Source

Next events on Google Open Source Live

Go Day

Nov 5, 9-11am PT

<https://goo.gle/GoDayNov5>

Golang experts will share updates on everything from **Go basics** to **Package Discovery** and **Editor Tooling**. Our partner, **Khan Academy** will also walk through an interesting use case about how the organization is using Go to save time and money.

Kubernetes Day

Dec 3, 9-11 am PT

<http://goo.gle/k8sDay>

At this event, Kubernetes experts at Google will cover the life of a **Kubernetes API**, **admission** webhooks, how **"apply"** works, and the distributed value store **etcd**.



Thank you!

We want to hear from you!
To discuss needs, use cases, or feedback reach out to
bryanzimmerman@google.com

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Source